

Shubham Darekar

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Computer Science undergraduate (2026) with production Python backend experience, published open-source libraries, and hands-on Linux systems administration.

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, Bash, C++

Python & Backend: FastAPI, REST API Design, Scalable Web Services, Automated Testing, Spring Boot, Node.js/Express

Web Front-End: React, Vite, React Router, Axios

Linux & Systems: Ubuntu/Debian, Shell Scripting, systemd, Package Management (apt), VPS Administration, Git

Containers & Cloud: Docker, Public Cloud (GCP – IAM, Hosting), Kubernetes (familiar), CI/CD (familiar)

Databases: MySQL, PostgreSQL, MongoDB, SQLite – SQL & NoSQL data modelling

Practices: Open-Source Contribution, Technical Documentation, Code Review, System Design (HLD/LLD), OOP, Data Structures & Algorithms

Experience

Cognizant Technology Solutions

GenC Next Intern – AI/ML & Data Science

Feb 2026 – May 2026

Pune, India

- Built and maintained ETL and data-warehouse pipelines processing 100K+ row datasets across 5+ configurations using **Python** and **SQL**, following modern, maintainable coding practices.
- Applied 6+ supervised and unsupervised ML algorithms with feature engineering to structured business datasets through an iterative development and validation workflow.
- Debugged and optimised SQL queries, reducing query complexity by 40% and improving pipeline execution performance through systematic troubleshooting.

Open Source & Linux

- Contributed bug fixes, documentation improvements, and small features to open-source repositories on **GitHub**, collaborating with maintainers through pull requests and code review.
- Administered personal **Ubuntu/Debian** VPS environments: configured **systemd** services and cron automation, and managed packages via **apt** for deployed personal projects.
- Wrote **Bash** scripts to automate deployment, environment setup, and routine maintenance across Linux development environments.
- Maintained and documented 2 published npm packages (*aAuth*, *alpha-sql*) with structured, automated test-case suites covering valid and invalid inputs – testing treated as a first-class part of the release process.
- Applied security-conscious design in *aAuth* – JWT access/refresh token rotation and BCrypt password hashing – for a reusable multitenant authentication service.

Projects

Kai – Multi-Agent AI Investment Analysis System

Python, FastAPI, React, PostgreSQL

- Architected a 4-agent **LLM** orchestration system (Coordinator, Fundamental, Sentiment, Valuation) with a structured debate loop, engineering agentic workflows across 2 LLM providers.
- Integrated **REST APIs** across 3 live data sources (Yahoo Finance, SEC EDGAR); built a full-stack FastAPI + React/Vite frontend with 5+ interactive dashboards.

HTRL – AI-Powered Real-Time Traffic Accident Monitoring

Python, YOLOv8, SQL, React

- Built an AI-native monitoring system using YOLOv8 across 10+ simulated CCTV feeds with confidence-threshold classification logic, achieving a <5% false-positive rate.
- Triggered automated dispatch to 3 emergency services via a React dashboard, with a full **SQL**-based event store for auditability.

Education

Dr. D. Y. Patil Institute of Technology

Bachelor of Engineering, Computer Science

Aug 2022 – May 2026

Pune, Maharashtra

Certifications

Database Management Systems (DBMS) – NPTEL, IIT, 2024 | Machine Learning, Feature Engineering, SQL Advanced, Python, ETL & Data Warehousing – Udemy Professional Certificates, 2024–2025